

Fault Hamiltonicity of Hierarchical Restricted HL-Graphs *

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Abstract

The hierarchical restricted HL-graph $HRHL(k, n)$ is a new kind of interconnection networks such as the hierarchical crossed cube $HCC(k, n)$. $HRHL(k, n)$ is constructed according to the hypercube and the restricted HL-graph. In this paper, we show that $HRHL(k, n)$ is $(k + n - 2)$ -fault hamiltonian and $(k + n - 3)$ -fault hamiltonian connected for $k \geq 1$ and $n \geq 3$.

Keywords: hierarchical crossed cube, hierarchical restricted HL-graphs, fault-tolerant, hamiltonian, hamiltonian connected.

1 Introduction

Recently, hierarchical interconnection networks have obtained lots of attention [3, 4, 10, 17]. An interconnection network is proposed and called *hierarchical crossed cube* (HCC) [7, 9, 11]. HCC draws upon constructions used within the well-known hypercube [14] and also the crossed cube [2]. The crossed cube is one member of restricted HL-graphs [16]. In this paper, we define the *hierarchical restricted HL-graphs* that HCC is one member of hierarchical restricted HL-graphs.

The embedding of linear arrays and rings into a faulty interconnection network is an important problem in parallel processing and it also relates to fault hamiltonicity of graph theory. There is a large of literature about fault hamiltonicity [5, 8, 18]. Since Park *et al.* [13] found that an n -dimensional restricted HL-graph is $(n - 2)$ -fault

hamiltonian and $(n - 3)$ -fault hamiltonian connected for $n \geq 3$, we are interested in investigating the fault hamiltonicity of hierarchical restricted HL-graphs in this paper.

The rest of this paper is organized as follows. In section 2, we provide definitions relative to the hierarchical restricted HL-graphs and necessary preliminaries. Then the main results are given in Section 3. Finally, the conclusions are given in Section 4.

2 Preliminary

For the graph definition and notation we follow [1, 6]. The vertices on the graph represent processors and edges represent links between processors. Let $G = (V, E)$ be an undirected graph where $V(G)$ is the vertex set and $E(G)$ is the edge set. Two vertices u and v of G are adjacent if $(u, v) \in E$.

A graph G is called *hamiltonian* if G contains a hamiltonian cycle. A graph G is called *hamiltonian connected* if there exists a hamiltonian path of G joining any two vertices [12]. A bipartite graph G is called *hamiltonian laceable* if there exists a hamiltonian path joining any two vertices which are in different partite sets [15]. A graph G is called *k -fault hamiltonian* if $G - F$ is hamiltonian for $F \subset V(G) \cup E(G)$ with $|F| \leq k$. A graph G is called *k -fault hamiltonian connected* if $G - F$ is hamiltonian connected for $F \subset V(G) \cup E(G)$ with $|F| \leq k$. A bipartite graph G is called *k -edge fault hamiltonian laceable* if $G - F$ is hamiltonian laceable for $F \subset E(G)$ with $|F| \leq k$.

Definition 1 [13] *A class of hypercube-like networks, called HL-graphs, which can be defined by applying the $\oplus$ operation repeatedly as follows: $HL_0 = \{K_1\}$; for $n \geq 1$, $HL_n = \{G_0 \oplus G_1 : G_0, G_1 \in HL_{n-1}\}$. Then,*

*This work was supported in part by the National Science Council of Republic of China under NSC 97-2221-E-259-020.

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- $HL_1 = \{K_2\}$,
- $HL_2 = \{C_4\}$,
- $HL_3 = \{Q_3, G(8, 4)\}$.

Here, C_4 is a cycle graph with 4 vertices, Q_3 is a 3-dimensional hypercube, and $G(8, 4)$ is a recursive circulant which is isomorphic to crossed cube CQ_3 and möbius ladder.

Definition 2 [13] A subclass of nonbipartite HL-graphs, called restricted HL-graphs, is defined recursively as follows:

- $RHL_n = HL_n$ for $0 \leq n \leq 2$;
- $RHL_3 = HL_3 - Q_3 = \{G(8, 4)\}$;
- $RHL_n = \{G_0 \oplus G_1 : G_0, G_1 \in RHL_{n-1}\}$ for $n \geq 4$.

Definition 3 Fix $k \geq 1, n \geq 1$. The hierarchical restricted HL-graph $HRHL(k, n)$ has vertex set $\{0, 1\}^{k+2n}$. Each vertex of $HRHL(k, n)$ is written as (u, v, w) , where $u \in \{0, 1\}^k$ and $v, w \in \{0, 1\}^n$ (throughout the paper, bold type denotes a bit-string). The set of edges of $HRHL(k, n)$ is partitioned into 2 sets, E_{int} and E_{ext} . The set E_{int} is referred to as the set of internal edges, whilst the set E_{ext} is referred to as the set of external edges. In more detail,

$E_{int} = \{((u, v, w), (u, v, w')) : (w, w') \text{ is an edge of } RHL_n\}$ and

$E_{ext} = \{((u, v, w), (u', w, v)) : (u, u') \text{ is an edge of } Q_k\}$.

In this paper, we only prove that $HCC(k, n)$ is $(k + n - 2)$ -fault hamiltonian and $(k + n - 3)$ -fault hamiltonian connected for $k \geq 1$ and $n \geq 3$. However, we can use the similar proof techniques to show that every member of $HRHL(k, n)$ is $(k + n - 2)$ -fault hamiltonian and $(k + n - 3)$ -hamiltonian connected for $k \geq 1$ and $n \geq 3$.

In effect, $HCC(k, n)$ is formed by taking 2^{k+n} disjoint copies of CQ_n , with $CQ_n(u, v)$ denoting the copy of CQ_n on the set of vertices $\{(u, v, w) : w \in \{0, 1\}^n\}$ (the edges of these copies of CQ_n form the internal edges). The vertices in these copies of CQ_n are then joined by additional edges (the external edges) whereby the vertices are partitioned into 2^{2n} sets of 2^k vertices, with each set of 2^k vertices joined by edges to form a copy of Q_k . Consequently, edges lie in the ‘internal layer’ or the ‘external layer’. Clearly, $HCC(k, n)$ has 2^{k+2n} vertices, $n2^{k+2n-1}$ internal edges, and $k2^{k+2n-1}$ external edge, making $(n + k)2^{k+2n-1}$ edges in total. Indeed, a simple induction yields that $HCC(k, n)$ is $(n + k)$ -regular.

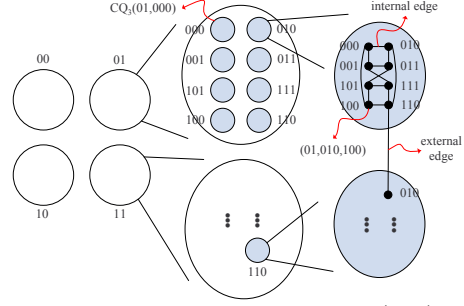


Figure 1: Structure of $HCC(2, 3)$

Given a graph $G = (V, E)$, let $F_v \subseteq V(G)$, $F_e \subseteq E(G)$ and $F = F_v \cup F_e$. Let $H = (V', E')$ be a subgraph of G . We write $F(H)$ to denote the set $(V' \cup E') \cap F$. In $HCC(k, n)$, let $F_{int} \subseteq E_{int}(HCC(k, n))$, $F_{ext} \subseteq E_{ext}(HCC(k, n))$, $F_e = F_{int} \cup F_{ext}$ and $F = F_v \cup F_e = F_v \cup F_{int} \cup F_{ext}$. We describe that the vertex u (or edge e) is fault free if u (or edge e) $\notin F$; and the vertex u (or edge e) is fault if u (or edge e) $\in F$.

We shall write a path of vertices in any graph as $\langle u = u_0 \rightarrow u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_m = v \rangle$, where $u_i \rightarrow u_{i+1}$ denotes that an edge joins u_i and u_{i+1} , or as $u \rightarrow^* v$ if we do not need to detail the vertices of the actual path. We also write the path $\langle u_0 \rightarrow u_1 \rightarrow u_2 \rightarrow u_3 \rightarrow \dots \rightarrow u_m \rangle$ as $\langle u_0 \rightarrow P(u_1, u_i) \rightarrow u_{i+1} \rightarrow \dots \rightarrow u_j \rightarrow P_1 \rightarrow u_k \rightarrow \dots \rightarrow u_m \rangle$, where $P(u_1, u_i)$ is the subpath $\langle u_1 \rightarrow u_2 \rightarrow \dots \rightarrow u_i \rangle$ and P_1 is the subpath $\langle u_j \rightarrow u_{j+1} \rightarrow \dots \rightarrow u_k \rangle$. The length of the path P , $l(P)$, is the number of edges on P . However, in $HCC(k, n)$ we write $(u, v, w) \rightarrow_{CQ_n} (u, v, w')$ to denote that the edge is an internal edge and $(u, v, w) \rightarrow_{Q_k} (u', w, v)$ to denote that the edge is an external edge. We write the path $\langle (u, v, w_1) \rightarrow_{CQ_n} (u, v, w_2) \rightarrow \dots \rightarrow_{CQ_n} (u, v, w_m) \rangle$ as $\langle (u, v, w_1) \rightarrow_{CQ_n}^* (u, v, w_m) \rangle$ if we do not need to detail the vertices of the actual path. We write $(u, v, w) \xrightarrow{H} (u, v, w')$ to denote a hamiltonian path of $CQ_n(u, v) - F(CQ_n(u, v))$ joining (u, v, w) and (u, v, w') . For any $u \in \{0, 1\}$, we denote by $\bar{u}$ the complementary bit to u , and we write $\mathbf{0}$ to denote a tuple of 0's (of some appropriate length). Finally, let $v_i, v_j \in \{0, 1\}^n$, $n \geq 1$, and let $v_i \neq v_j$ for $i \neq j$ in the following discussions.

Lemma 1 [13] Every n -dimensional restricted HL-graph, $n \geq 3$, is $(n - 3)$ -fault hamiltonian connected and $(n - 2)$ -fault hamiltonian.

For two distinct vertices (u, v, w) and (u, v', w) of $HCC(1, n)$, $n \geq 3$, since CQ_n is hamiltonian connected, there exists a hamiltonian path of $CQ_n(\bar{u}, w)$ joining $(\bar{u}, w, v)$ and $(\bar{u}, w, v')$. Hence, we have Observation 1 and a path

as $P((u, v, w), (u, v', w)) = \langle (u, v, w) \rightarrow_{Q_1} (\bar{u}, w, v) \rightarrow_{CQ_n}^H (\bar{u}, w, v') \rightarrow_{Q_1} (u, v', w) \rangle$. (see fig.2(a))

Observation 1 For any two vertices (u, v, w) and (u, v', w) of $HCC(1, n)$, $n \geq 3$, there exists a path $P((u, v, w), (u, v', w))$ such that it traverses every vertex of $V(CQ_n(\bar{u}, w)) \cup \{(u, v, w), (u, v', w)\}$ exactly once.

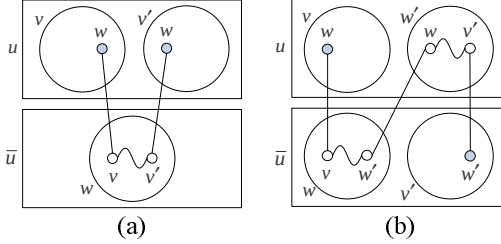


Figure 2: Illustrations of Observation 1 and Observation 2

Let (u, v, w) and $(\bar{u}, v', w')$ be two distinct vertices of $HCC(1, n)$ with $v \neq w'$ and $v' \neq w$ for $n \geq 3$. Clearly, there exist a hamiltonian path of $CQ_n(\bar{u}, w)$ joining $(\bar{u}, w, v)$ and $(\bar{u}, w, w')$ and a hamiltonian path of $CQ_n(u, w')$ joining (u, w', w) and (u, w', v') . Hence, we have Observation 2 and a path as $P((u, v, w), (\bar{u}, v', w')) = \langle (u, v, w) \rightarrow_{Q_1} (\bar{u}, w, v) \rightarrow_{CQ_n}^H (\bar{u}, w, w') \rightarrow_{Q_1} (u, w', w) \rightarrow_{CQ_n}^H (u, w', v') \rightarrow_{Q_1} (\bar{u}, v', w') \rangle$. (see fig.2(b))

Observation 2 For any two vertices (u, v, w) and $(\bar{u}, v', w')$ of $HCC(1, n)$ with $v \neq w'$ and $v' \neq w$ for $n \geq 3$, there exists a path $P((u, v, w), (\bar{u}, v', w'))$ such that it traverses every vertex of $V(CQ_n(\bar{u}, w)) \cup V(CQ_n(u, w')) \cup \{(u, v, w), (\bar{u}, v', w')\}$ exactly once.

Lemma 2 The complete bipartite graph $K_{3,3}$ is 1-edge fault hamiltonian laceable.

Proof. Let $V(K_{3,3}) = \{00, 01, 02, 10, 11, 12\}$, and let $\{00, 01, 02\}$ and $\{10, 11, 12\}$ be two partite sets of $K_{3,3}$. Without loss of generality, assume $F = \{(00, 10)\}$. (see fig.3(a)) Then the corresponding hamiltonian paths are listed below. $\square$

$F = \{(00, 10)\}$	
$\langle 00, 11, 01, 12, 02, 10 \rangle$	$\langle 00, 12, 01, 10, 02, 11 \rangle$
$\langle 00, 11, 01, 10, 02, 12 \rangle$	$\langle 01, 11, 00, 12, 02, 10 \rangle$
$\langle 01, 10, 02, 12, 00, 11 \rangle$	$\langle 01, 10, 02, 11, 00, 12 \rangle$
$\langle 02, 11, 00, 12, 01, 10 \rangle$	$\langle 02, 10, 01, 12, 00, 11 \rangle$
$\langle 02, 10, 01, 11, 00, 12 \rangle$	

Lemma 3 The complete bipartite graph $K_{t,t}$, $t \geq 3$, is $(t-2)$ -edge fault hamiltonian laceable.

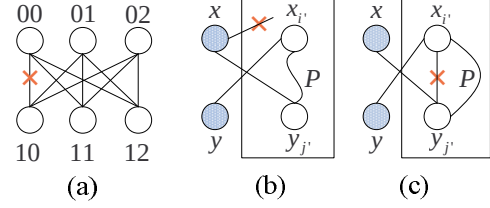


Figure 3: Illustrations of Lemma 2 and Lemma 3

Proof. We prove this lemma by induction on t . It follows from Lemma 2 that $K_{3,3}$ is 1-edge fault hamiltonian laceable. Suppose that $K_{t-1,t-1}$ is $(t-3)$ -edge fault hamiltonian laceable for $t \geq 4$. Then we prove that $K_{t,t}$ is $(t-2)$ -edge fault hamiltonian laceable. Let $X = \{x_i : i = 1, 2, \dots, t\}$, $Y = \{y_j : j = 1, 2, \dots, t\}$ be two partite sets and $V(K_{t,t}) = X \cup Y$. Let F be a fault set of $E(K_{t,t})$ with $|F| \leq t-2$ and let $x \in X$ and $y \in Y$. We want to construct a hamiltonian path joining x and y on $K_{t,t} - F$.

Case 1: There exists some edge (x, y_j) or $(y, x_i) \in F$, $1 \leq i, j \leq t$.

Let $X' = X - \{x\}$, $Y' = Y - \{y\}$ and let $V(K_{t-1,t-1}) = X' \cup Y'$. Since there exists some edge (x, y_j) or $(y, x_i) \in F$, $1 \leq i, j \leq t$, $|F(K_{t-1,t-1})| \leq t-3$. There exist some vertices $x_{i'} \in X'$ and $y_{j'} \in Y'$ of $K_{t-1,t-1}$ that the edges $(x, y_{j'})$ and $(y, x_{i'})$ are fault free, $1 \leq i', j' \leq t$. (see fig.3(b)) Since $K_{t-1,t-1}$ is $(t-3)$ -edge fault hamiltonian laceable, there exists a hamiltonian path P of $K_{t-1,t-1} - F(K_{t-1,t-1})$ joining $x_{i'}$ and $y_{j'}$. Then $\langle x \rightarrow y_{j'} \rightarrow P \rightarrow x_{i'} \rightarrow y \rangle$ forms a hamiltonian path of $K_{t,t} - F$ joining x and y .

Case 2: All edges in $\{(x, y_j) : 1 \leq j \leq t\} \cup \{(y, x_i) : 1 \leq i \leq t\}$ are fault free.

Let $X' = X - \{x\}$, $Y' = Y - \{y\}$ and let $V(K_{t-1,t-1}) = X' \cup Y'$. Clearly, there exists some edge $(x_{i'}, y_{j'}) \in F$, $1 \leq i', j' \leq t$, where $x_{i'} \in X'$ and $y_{j'} \in Y'$. (see fig.3(c)) Since $K_{t-1,t-1}$ is $(t-3)$ -edge fault hamiltonian laceable, it is not difficult to verify that there exists a hamiltonian path P of $K_{t-1,t-1} - (F(K_{t-1,t-1}) - \{(x_{i'}, y_{j'})\})$ joining $x_{i'}$ and $y_{j'}$. Then $\langle x \rightarrow y_{j'} \rightarrow P \rightarrow x_{i'} \rightarrow y \rangle$ forms a hamiltonian path of $K_{t,t} - F$ joining x and y . $\square$

To simplify the explanation, if no ambiguity arises, consider each subgraph $CQ_n(u, v)$, $u \in \{0, 1\}^k$, $v \in \{0, 1\}^n$, as a super vertex, and consider $HCC(1, n)$ as a complete bipartite graph $K_{2^n, 2^n}$ of super vertices. Herein, the super vertex of $CQ_n(u, v)$ is labeled by uv . A hamiltonian path consists of super vertices of $HCC(1, n)$ is called a *super hamiltonian path*. Then we have Lemma 4 on super hamiltonian path.

Lemma 4 *Let $4 \leq t \leq 2^n, 1 \leq i, j \leq t, p_i \in \{0, 1\}^n, q_j \in \{0, 1\}^n$, and $u \in \{0, 1\}$. Let H be the subgraph of $HCC(1, n), n \geq 3$, induced by $2 \times t \times 2^n$ vertices of $CQ_n(u, p_i) \cup CQ_n(\bar{u}, q_j)$. Let F be the fault set of H and let (u, v, w) and $(\bar{u}, v', w')$ be two fault free vertices in H . There exists a hamiltonian path of $H - F$ joining (u, v, w) and $(\bar{u}, v', w')$ if $|F| \leq t - 4$, $|F(CQ_n(u, p_i))| \leq n - 3$, and $|F(CQ_n(\bar{u}, q_i))| \leq n - 3$ for all $1 \leq i, j \leq t$.*

Proof. Let $F' = \{e : e \text{ is the external edge which is incident to the vertex of } F_v \cup \{(u, v, w), (\bar{u}, v', w')\}\}$. Since every vertex of $HCC(1, n)$ is incident to only one external edge, $|F'| \leq |F_v| + 2$. Therefore, $|F_{ext} \cup F'| \leq |F_{ext} \cup F_v| + 2 \leq |F| + 2 \leq t - 2$. By Lemma 3, there exists a super hamiltonian path P of $H - (F_{ext} \cup F')$ joining uv and $\bar{u}v'$. Let $v = p_1$ and $v' = q_t$ and let $P = \langle uv = up_1 \rightarrow \bar{u}q_1 \rightarrow up_2 \rightarrow \bar{u}q_2 \dots \rightarrow up_t \rightarrow \bar{u}q_t = \bar{u}v' \rangle$. (see fig.4) Clearly, there exists a hamiltonian path of $CQ_n(u, p_i) - F(CQ_n(u, p_i))$ joining (u, p_i, q_{i-1}) and (u, p_i, q_i) for each $1 \leq i \leq t$ (if $i = 1$, substitute w for q_{i-1}). Similarly, there exists a hamiltonian path of $CQ_n(\bar{u}, q_j) - F(CQ_n(\bar{u}, q_j))$ joining $(\bar{u}, q_j, p_j)$ and $(\bar{u}, q_j, p_{j+1})$ for each $1 \leq j \leq t$ (if $j = t$, substitute w' for p_{j+1}). Let $P((u, v, w), (\bar{u}, v', w')) = \langle (u, v, w) = (u, p_1, w) \xrightarrow{H} CQ_n(u, p_1, q_1) \rightarrow_{Q_1} (\bar{u}, q_1, p_1) \xrightarrow{H} CQ_n(\bar{u}, q_1, p_2) \rightarrow_{Q_1} (u, p_2, q_1) \xrightarrow{H} CQ_n(u, p_2, q_2) \dots \rightarrow_{Q_1} (\bar{u}, q_t, p_t) \xrightarrow{H} CQ_n(\bar{u}, q_t, w') = (\bar{u}, v', w') \rangle$. Then $P((u, v, w), (\bar{u}, v', w'))$ forms a hamiltonian path of $H - F(H)$ joining (u, v, w) and $(\bar{u}, v', w')$. $\square$

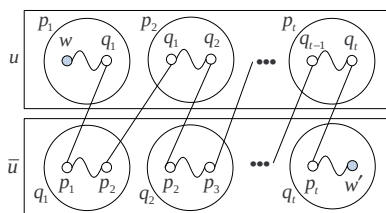


Figure 4: Illustration of Lemma 4

3 Main result

Lemma 5 $HCC(1, n)$ is $(n-2)$ -fault hamiltonian connected for $n > 3$.

Proof. Let $x = (u, v, w)$ and $y = (u', v', w')$ be two fault free vertices of $HCC(k, n)$ and let F be the fault set of $HCC(1, n)$ with $|F| \leq n - 2$. Without lost of generality, assume that $|F(CQ_n(0, \mathbf{0}))| \geq |F(CQ_n(i, j))|, i \in \{0, 1\}, j \in \{0, 1\}^n$. Then consider F with two cases: (1)

$$|F(CQ_n(0, \mathbf{0}))| = n - 2 \text{ and (2) } |F(CQ_n(i, j))| \leq n - 3 \text{ for each } CQ_n(i, j).$$

Case 1: $|F(CQ_n(0, \mathbf{0}))| = n - 2$.

Clearly, $F = F(CQ_n(0, \mathbf{0}))$. Therefore, every $CQ_n(i, j)$ of $HCC(1, n)$ except $CQ_n(0, \mathbf{0})$ is hamiltonian connected. Without loss of generality, assume $x \in V(CQ_n(0, \mathbf{0}))$ if x or $y \in V(CQ_n(0, \mathbf{0}))$ and assume $u = 0, u' = 1$ if $u \neq u'$. We consider the relative positions of x and y as eight conditions (see Table 1).

$u = u'$	$v = v'$	$x, y \in V(CQ_n(0, \mathbf{0}))$	subcase1.1
		$x, y \notin V(CQ_n(0, \mathbf{0})), u = u' = 0$	subcase1.2
		$u = u' = 1$	subcase1.3
	$v \neq v'$	$x \in V(CQ_n(0, \mathbf{0}))$	subcase1.4
		$x \notin V(CQ_n(0, \mathbf{0})), u = u' = 0$	subcase1.5
		$u = u' = 1$	subcase1.6
$u \neq u'$ $(u = 0, u' = 1)$	$x \in V(CQ_n(0, \mathbf{0}))$	subcase1.7	
	$x \notin V(CQ_n(0, \mathbf{0}))$	subcase1.8	

Table 1: The relative positions of x and y for Case 1 of Lemma 5

Subcase 1.1: $x = (0, \mathbf{0}, w)$, $y = (0, \mathbf{0}, w')$, $w \neq w'$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. We consider C with two cases: (1) Suppose $(0, \mathbf{0}, w)$ and $(0, \mathbf{0}, w')$ are adjacent on C . Let $(0, \mathbf{0}, w), (0, \mathbf{0}, w_1)$, and $(0, \mathbf{0}, w_2)$ be three successive vertices on C and let $C = \langle (0, \mathbf{0}, w) \rightarrow_{CQ_n} (0, \mathbf{0}, w_1) \rightarrow_{CQ_n} (0, \mathbf{0}, w_2) \rightarrow_{CQ_n}^* (0, \mathbf{0}, w') \rightarrow_{CQ_n} (0, \mathbf{0}, w) \rangle$. (see fig.5(a)) (2) Suppose $(0, \mathbf{0}, w)$ and $(0, \mathbf{0}, w')$ are not adjacent on C . Let $C = \langle (0, \mathbf{0}, w) \rightarrow_{CQ_n}^* (0, \mathbf{0}, w_1) \rightarrow_{CQ_n} (0, \mathbf{0}, w') \rightarrow_{CQ_n}^* (0, \mathbf{0}, w_2) \rightarrow_{CQ_n} (0, \mathbf{0}, w) \rangle$. (see fig.5(b)) Let $(1, w_1, w_3)$ be a vertex of $CQ_n(1, w_1)$ with $w_3 \neq \mathbf{0}$ and let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(1, w_1))$. (see fig.5(c)) By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((0, w_3, w_1), (1, w_2, \mathbf{0}))$ in H . Let $P(x, y) = \langle x = (0, \mathbf{0}, w) \rightarrow_{CQ_n}^* (0, \mathbf{0}, w_1) \rightarrow_{Q_1} (1, w_1, \mathbf{0}) \rightarrow_{CQ_n}^H (1, w_1, w_3) \rightarrow_{Q_1} P((0, w_3, w_1), (1, w_2, \mathbf{0})) \rightarrow_{Q_1} (0, \mathbf{0}, w_2) \rightarrow_{CQ_n}^* (0, \mathbf{0}, w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.2: $x = (0, v, w), y = (0, v, w'), v \neq 0, w \neq w'$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Clearly, there exists a hamiltonian path P of $CQ_n(0, v)$ joining $(0, v, w)$ and $(0, v, w')$. Assume $P = \langle (0, v, w) \rightarrow_{CQ_n}^* (0, v, w_1) \rightarrow_{CQ_n} (0, v, w_2) \rightarrow_{CQ_n}^* (0, v, w') \rangle$ with $(0, \mathbf{0}, w_1) \in C$ and assume $C = \langle (0, \mathbf{0}, w_1) \rightarrow_{CQ_n}^H (0, \mathbf{0}, w_3) \rightarrow_{CQ_n} (0, \mathbf{0}, w_1) \rangle$. (see fig.5(d)) By Observation 1, there exists a

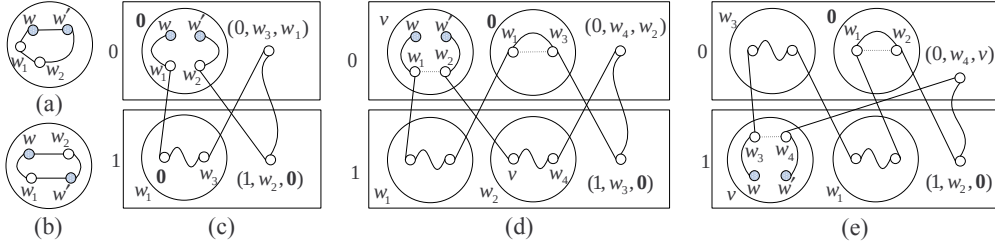


Figure 5: Illustrations of Subcase 1.1, Subcase 1.2, and Subcase 1.3 in Lemma 5

path $P((0, v, w_1), (0, \mathbf{0}, w_1))$ that it traverses every vertex of $V(CQ_n(1, w_1)) \cup \{(0, v, w_1), (0, \mathbf{0}, w_1)\}$ exactly once. Let $(1, w_2, w_4)$ be a vertex of $CQ_n(1, w_2)$ with $w_4 \neq v$ and $w_4 \neq \mathbf{0}$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, v) \cup CQ_n(1, w_1) \cup CQ_n(1, w_2))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_3, \mathbf{0}), (0, w_4, w_2))$ in H . Let $P(x, y) = \langle x = (0, v, w) \xrightarrow{*CQ_n} P((0, v, w_1), (0, \mathbf{0}, w_1)) \xrightarrow{H} CQ_n(0, \mathbf{0}, w_3) \xrightarrow{Q_1} P((1, w_3, \mathbf{0}), (0, w_4, w_2)) \xrightarrow{Q_1} (1, w_2, w_4) \xrightarrow{H} CQ_n(1, w_2, v) \xrightarrow{Q_1} (0, v, w_2) \xrightarrow{*CQ_n} (0, v, w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.3: $x = (1, v, w), y = (1, v, w'), w \neq w'$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be two successive vertices of C with $w_1 \neq v$ and $w_2 \neq v$. (see fig.5(e)) Clearly, there exists a hamiltonian path P of $CQ_n(1, v)$ joining $(1, v, w)$ and $(1, v, w')$. Assume that $P = \langle (1, v, w) \xrightarrow{*CQ_n} (1, v, w_3) \xrightarrow{CQ_n} (1, v, w_4) \xrightarrow{*CQ_n} (1, v, w') \rangle$ where $w_3 \neq \mathbf{0}$ and $w_4 \neq \mathbf{0}$. By Observation 2, since $w_3 \neq \mathbf{0}$ and $w_1 \neq v$, there exists a path $P((1, v, w_3), (0, \mathbf{0}, w_1))$ that it traverses every vertex of $V(CQ_n(0, w_3)) \cup V(CQ_n(1, w_1)) \cup \{(1, v, w_3), (0, \mathbf{0}, w_1)\}$ exactly once. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, w_3) \cup CQ_n(1, v) \cup CQ_n(1, w_1))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_2, \mathbf{0}), (0, w_4, v))$ in H . Let $P(x, y) = \langle x = (1, v, w) \xrightarrow{*CQ_n} P((1, v, w_3), (0, \mathbf{0}, w_1)) \xrightarrow{H} CQ_n(0, \mathbf{0}, w_2) \xrightarrow{Q_1} P((1, w_2, \mathbf{0}), (0, w_4, v)) \xrightarrow{Q_1} (1, v, w_4) \xrightarrow{*CQ_n} (1, v, w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.4: $x = (0, \mathbf{0}, w), y = (0, v', w'), v' \neq \mathbf{0}$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ be a neighbor of $(0, \mathbf{0}, w)$ on C with $w_1 \neq w'$. (see fig.6(a)) Clearly, there exists a hamiltonian

path P of $CQ_n(0, v')$ joining $(0, v', w_1)$ and $(0, v', w')$. Assume that $P = \langle (0, v', w_1) \xrightarrow{*CQ_n} (0, v', w_2) \xrightarrow{CQ_n} (0, v', w_3) \xrightarrow{*CQ_n} (0, v', w') \rangle$. By Observation 1, there exists a path $P((0, \mathbf{0}, w_1), (0, v', w_1))$ that it traverses every vertex of $V(CQ_n(1, w_1)) \cup \{(0, \mathbf{0}, w_1), (0, v', w_1)\}$ exactly once. Let $(1, w_2, w_4)$ be a vertex of $CQ_n(1, w_2)$ with $w_4 \neq v'$ and $w_4 \neq \mathbf{0}$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, v') \cup CQ_n(1, w_1) \cup CQ_n(1, w_2))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((0, w_4, w_2), (1, w_3, v'))$ in H . Let $P(x, y) = \langle x = (0, \mathbf{0}, w) \xrightarrow{H} CQ_n P((0, \mathbf{0}, w_1), (0, v', w_1)) \xrightarrow{*CQ_n} (0, v', w_2) \xrightarrow{Q_1} (1, w_2, v') \xrightarrow{H} CQ_n(1, w_2, w_4) \xrightarrow{Q_1} P((0, w_4, w_2), (1, w_3, v')) \xrightarrow{Q_1} (0, v', w_3) \xrightarrow{*CQ_n} (0, v', w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.5: $x = (0, v, w), y = (0, v', w'), v \neq v', v \neq \mathbf{0}$, and $v' \neq \mathbf{0}$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be two successive vertices of C with $w_1 \neq w$ and $w_2 \neq w'$. (see fig.6(b)) Assume that $C = \langle (0, \mathbf{0}, w_1) \xrightarrow{*CQ_n} (0, \mathbf{0}, w_3) \xrightarrow{CQ_n} (0, \mathbf{0}, w_4) \xrightarrow{*CQ_n} (0, \mathbf{0}, w_2) \xrightarrow{CQ_n} (0, \mathbf{0}, w_1) \rangle$. By Observation 1, there exists a path $P((0, v, w_1), (0, \mathbf{0}, w_1))$ that it traverses every vertex of $V(CQ_n(1, w_1)) \cup \{(0, v, w_1), (0, \mathbf{0}, w_1)\}$ exactly once. Similarly, there exists a path $P((0, \mathbf{0}, w_2), (0, v', w_2))$ that it traverses every vertex of $V(CQ_n(1, w_2)) \cup \{(0, \mathbf{0}, w_2), (0, v', w_2)\}$ exactly once. Let $(1, w_4, w_5)$ be a vertex of $CQ_n(1, w_4)$ with $w_5 \neq v, w_5 \neq v'$, and $w_5 \neq \mathbf{0}$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, v) \cup CQ_n(0, v') \cup CQ_n(1, w_1) \cup CQ_n(1, w_2) \cup CQ_n(1, w_4))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_3, \mathbf{0}), (0, w_5, w_4))$ in H . Let $P(x, y) = \langle x = (0, v, w) \xrightarrow{H} CQ_n P((0, v, w_1), (0, \mathbf{0}, w_1)) \xrightarrow{*CQ_n} (0, \mathbf{0}, w_3) \xrightarrow{Q_1} P((1, w_3, \mathbf{0}), (0, w_5, w_4)) \xrightarrow{Q_1} (1, w_4, w_5) \xrightarrow{H} CQ_n(1, w_4, \mathbf{0}) \xrightarrow{Q_1} (0, \mathbf{0}, w_4) \xrightarrow{*CQ_n} P((0, \mathbf{0}, w_2), (0, v', w_2)) \xrightarrow{H} CQ_n(0, v', w') = y \rangle$.

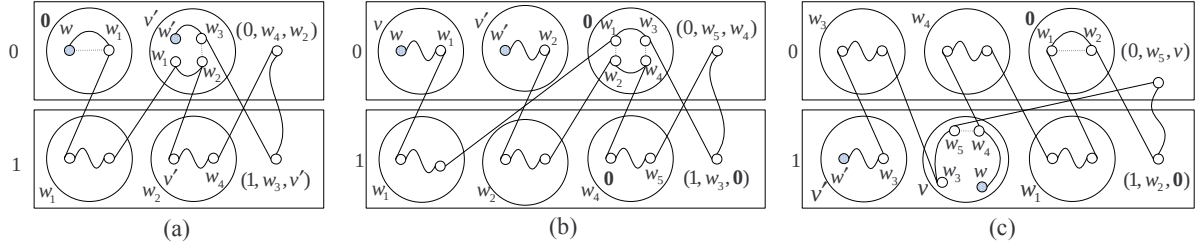


Figure 6: Illustrations of Subcase 1.4, Subcase 1.5, and Subcase 1.6 in Lemma 5

Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.6: $x = (1, v, w), y = (1, v', w'), v \neq v'$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be two successive vertices of C with $w_1, w_2 \neq v$ and $w_1, w_2 \neq v'$. (see fig.6(c)) Clearly, there exists a hamiltonian path P of $CQ_n(1, v)$ joining $(1, v, w)$ and some $(1, v, w_3)$ with $w_3 \neq w'$ and $w_3 \neq \mathbf{0}$. Assume that $P = \langle (1, v, w) \xrightarrow{*CQ_n} (1, v, w_4) \xrightarrow{CQ_n} (1, v, w_5) \xrightarrow{*CQ_n} (1, v, w_3) \rangle$ where $w_4 \neq \mathbf{0}$ and $w_5 \neq \mathbf{0}$. By Observation 2, since $w_4 \neq \mathbf{0}$ and $w_1 \neq v$, there exists a path $P((1, v, w_4), (0, \mathbf{0}, w_1))$ that it traverses every vertex of $V(CQ_n(0, w_4)) \cup V(CQ_n(1, w_1)) \cup \{(1, v, w_4), (0, \mathbf{0}, w_1)\}$ exactly once. By Observation 1, there exists a path $P((1, v, w_3), (1, v', w_3))$ that it traverses every vertex of $V(CQ_n(0, w_3)) \cup \{(1, v, w_3), (1, v', w_3)\}$ exactly once. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, w_3) \cup CQ_n(0, w_4) \cup CQ_n(1, v) \cup CQ_n(1, v') \cup CQ_n(1, w_1))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_2, \mathbf{0}), (0, w_5, v))$ in H . Let $P(x, y) = \langle x = (1, v, w) \xrightarrow{*CQ_n} P((1, v, w_4), (0, \mathbf{0}, w_1)) \xrightarrow{H} (0, \mathbf{0}, w_2) \xrightarrow{Q_1} P((1, w_2, \mathbf{0}), (0, w_5, v)) \xrightarrow{Q_1} (1, v, w_5) \xrightarrow{*CQ_n} P((1, v, w_3), (1, v', w_3)) \xrightarrow{H} (1, v', w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 1.7: $x = (0, \mathbf{0}, w), y = (1, v', w')$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ be a neighbor of $(0, \mathbf{0}, w)$ on C with $w_1 \neq v'$. (see fig.7(a)) Clearly, there exists a hamiltonian path of $CQ_n(1, v')$ joining $(1, v', w')$ and some $(1, v', w_2)$ with $w_2 \neq \mathbf{0}$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(1, v'))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_1, \mathbf{0}), (0, w_2, v'))$ in H . Let $P(x, y) = \langle x = (0, \mathbf{0}, w) \xrightarrow{H} (0, \mathbf{0}, w_1) \xrightarrow{Q_1} P((1, w_1, \mathbf{0}), (0, w_2, v')) \xrightarrow{Q_1} (1, v', w_2) \xrightarrow{H} (1, v', w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian

path of $HCC(1, n) - F$ joining x and y .

Subcase 1.8: $x = (0, v, w), y = (1, v', w'), v \neq \mathbf{0}$.

By Lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - F$. Let $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be two successive vertices of C with $w_1 \neq w, w_1 \neq v'$, and $w_2 \neq v'$. (see fig.7(b)) By Observation 1, there exists a path $P((0, v, w_1), (0, \mathbf{0}, w_1))$ that it traverses every vertex of $V(CQ_n(1, w_1)) \cup \{(0, v, w_1), (0, \mathbf{0}, w_1)\}$ exactly once. Let $(1, v', w_3)$ be a vertex of $CQ_n(1, v')$ with $w_3 \neq \mathbf{0}$ and $w_3 \neq v$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(0, v) \cup CQ_n(1, v') \cup CQ_n(1, w_1))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_2, \mathbf{0}), (0, w_3, v'))$ in H . Let $P(x, y) = \langle x = (0, v, w) \xrightarrow{H} CQ_n P((0, v, w_1), (0, \mathbf{0}, w_1)) \xrightarrow{H} (0, \mathbf{0}, w_2) \xrightarrow{Q_1} P((1, w_2, \mathbf{0}), (0, w_3, v')) \xrightarrow{Q_1} (1, v', w_3) \xrightarrow{H} (1, v', w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

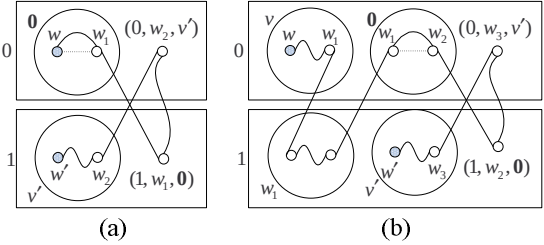


Figure 7: Illustrations of Subcase 1.7 and Subcase 1.8 in Lemma 5

Case2: $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$.

Hence, $CQ_n(i, j) - F(CQ_n(i, j))$ is hamiltonian connected. Next, consider the following three subcases.

Subcase 2.1: $x = (u, v, w), y = (u, v, w'), w \neq w'$.

Clearly, there exists a hamiltonian path P of $CQ_n(u, v) - F(CQ_n(u, v))$ joining (u, v, w) and (u, v, w') . Since $\lfloor l(P)/2 \rfloor > |F - F(CQ_n(u, v))|$, there exists two successive vertices (u, v, w_1) and (u, v, w_2) on P such that the ex-

ternal edges $((u, v, w_1), (\bar{u}, w_1, v))$ and $((u, v, w_2), (\bar{u}, w_2, v))$ are both fault free. We can write P as $\langle (u, v, w) \rightarrow_{CQ_n}^* (u, v, w_1) \rightarrow_{CQ_n} (u, v, w_2) \rightarrow_{CQ_n}^* (u, v, w') \rangle$. (see fig.8(a)) Let $((\bar{u}, w_1, w_3), (u, w_3, w_1))$ be a fault free external edge with $w_3 \neq v$ and let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(u, v) \cup CQ_n(\bar{u}, w_1))$. Let $t = 2^n - 1$. By Lemma 4, since $|F(H)| \leq n - 2 \leq t - 4 = 2^n - 5$ and $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$, there exists a hamiltonian path $P((u, w_3, w_1), (\bar{u}, w_2, v))$ in $H - F(H)$. Let $P(x, y) = \langle x = (u, v, w) \rightarrow_{CQ_n}^* (u, v, w_1) \rightarrow_{Q_1} (\bar{u}, w_1, v) \rightarrow_{CQ_n}^H (\bar{u}, w_1, w_3) \rightarrow_{Q_1} P((u, w_3, w_1), (\bar{u}, w_2, v)) \rightarrow_{Q_1} (u, v, w_2) \rightarrow_{CQ_n}^* (u, v, w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 2.2: $x = (u, v, w), y = (u, v', w'), v \neq v'$.

Since $|F| \leq n - 2 < 2^n - 2$, there exists the binary string $w_1 \in \{0, 1\}^n, w_1 \neq w$ and $w_1 \neq w'$, such that the external edges $((u, v, w_1), (\bar{u}, w_1, v))$ and $((u, v', w_1), (\bar{u}, w_1, v'))$ are both fault free. (see fig.8(b)) Clearly, there exists a hamiltonian path P of $CQ_n(u, v') - F(CQ_n(u, v'))$ joining (u, v', w_1) and (u, v', w') . Since $\lfloor l(P)/2 \rfloor > |F - F(CQ_n(u, v'))|$, there exists two successive vertices (u, v', w_2) and (u, v', w_3) of P such that the external edges $((u, v', w_2), (\bar{u}, w_2, v'))$ and $((u, v', w_3), (\bar{u}, w_3, v'))$ are both fault free. We can write P as $\langle (u, v', w_1) \rightarrow_{CQ_n}^* (u, v', w_2) \rightarrow_{CQ_n} (u, v', w_3) \rightarrow_{CQ_n}^* (u, v', w') \rangle$. Let $((\bar{u}, w_2, w_4), (u, w_4, w_2))$ be a fault free external edge with $w_4 \neq v$ and $w_4 \neq v'$. Let H be the subgraph of $HCC(1, n)$ induced by vertices of $HCC(1, n) - (CQ_n(u, v) \cup CQ_n(u, v') \cup CQ_n(\bar{u}, w_1) \cup CQ_n(\bar{u}, w_2))$. Let $t = 2^n - 2$. By Lemma 4, since $|F(H)| \leq n - 2 \leq t - 4 = 2^n - 6$ and $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$, there exists a hamiltonian path $P((u, w_4, w_2), (\bar{u}, w_3, v'))$ in $H - F(H)$. Let $P(x, y) = \langle x = (u, v, w) \rightarrow_{CQ_n}^H (u, v, w_1) \rightarrow_{Q_1} (\bar{u}, w_1, v) \rightarrow_{CQ_n}^H (\bar{u}, w_1, v') \rightarrow_{Q_1} (u, v', w_1) \rightarrow_{CQ_n}^* (u, v', w_2) \rightarrow_{Q_1} (\bar{u}, w_2, v') \rightarrow_{CQ_n}^H (\bar{u}, w_2, w_4) \rightarrow_{Q_1} P((u, w_4, w_2), (\bar{u}, w_3, v')) \rightarrow_{Q_1} (u, v', w_3) \rightarrow_{CQ_n}^* (u, v', w') = y \rangle$. Then $P(x, y)$ forms a hamiltonian path of $HCC(1, n) - F$ joining x and y .

Subcase 2.3: $x = (u, v, w), y = (\bar{u}, v', w')$.

Let $t = 2^n$. By Lemma 4, since $|F| \leq n - 2 \leq t - 4 = 2^n - 4$ and $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$, there exists a hamiltonian path of $HCC(1, n) - F$ joining x and y .

This lemma is proved. $\square$

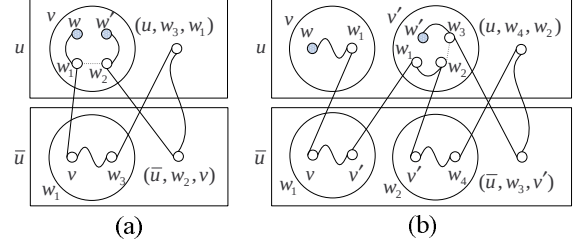


Figure 8: Illustrations of Subcase 2.1 and Subcase 2.2 in Lemma 5

Lemma 6 $HCC(1, n)$ is $(n-1)$ -fault hamiltonian for $n \geq 3$.

Proof. Without loss of generality, let $|F(CQ_n(0, \mathbf{0}))| \geq |F(CQ_n(i, j))|, i \in \{0, 1\}, j \in \{0, 1\}^n$. Consider F with three cases: (1) $|F(CQ_n(0, \mathbf{0}))| = n - 1$, (2) $|F(CQ_n(0, \mathbf{0}))| = n - 2$, and (3) $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$.

Case 1: $|F(CQ_n(0, \mathbf{0}))| = n - 1$.

Let $f_1 \in F(CQ_n(0, \mathbf{0}))$. By lemma 1, there exists a hamiltonian cycle C in $CQ_n(0, \mathbf{0}) - (F(CQ_n(0, \mathbf{0})) - f_1)$. We consider f_1 with two cases: (1) Suppose $f_1 \in C$. Let two vertices $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be the neighbors of f_1 on C . Hence, there exists a hamiltonian path of $CQ_n(0, \mathbf{0}) - F(CQ_n(0, \mathbf{0}))$ joining $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$. (2) Suppose $f_1 \notin C$. Let $C = \langle (0, \mathbf{0}, w_1) \rightarrow_{CQ_n}^H (0, \mathbf{0}, w_2) \rightarrow_{CQ_n} (0, \mathbf{0}, w_1) \rangle$. (see fig.9(a)) Then let $(1, w_2, w_3)$ be a vertex of $CQ_n(1, w_2)$ with $w_3 \neq \mathbf{0}$ and let H be the subgraph of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(1, w_2))$. By lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((0, w_3, w_2), (1, w_1, \mathbf{0}))$ in H . Let $C = \langle (0, \mathbf{0}, w_1) \rightarrow_{CQ_n}^H (0, \mathbf{0}, w_2) \rightarrow_{Q_1} (1, w_2, \mathbf{0}) \rightarrow_{CQ_n}^H (1, w_2, w_3) \rightarrow_{Q_1} P((0, w_3, w_2), (1, w_1, \mathbf{0})) \rightarrow_{Q_1} (0, \mathbf{0}, w_1) \rangle$. Then C forms a hamiltonian cycle of $HCC(1, n) - F$.

Case 2: $|F(CQ_n(0, \mathbf{0}))| = n - 2$.

Hence, $|F - F(CQ_n(0, \mathbf{0}))| = 1$. Next, consider the following two subcases.

Subcase 2.1: $n = 3$ and $F = F_v \cup F_{int}$.

Hence, $|F| = |F_v \cup F_{int}| = 2$ and $|F(CQ_3(0, \mathbf{0}))| = 1$. Let the other fault is in $CQ_3(u, v)$. Clearly, every $CQ_3(i, j)$ of $HCC(1, 3)$ except $CQ_3(0, \mathbf{0})$ and $CQ_3(u, v)$ is hamiltonian connected.

subcase 2.1.1: $u = 0$.

It is not difficult to verify that there exists two vertices $(0, \mathbf{0}, w_1)$ and $(0, v, w_1) \notin F$. Since CQ_3 is 1-fault hamiltonian, there exists a hamiltonian cycle C_1 in $CQ_3(0, \mathbf{0}) - F(CQ_3(0, \mathbf{0}))$ and a hamiltonian cycle C_2 in

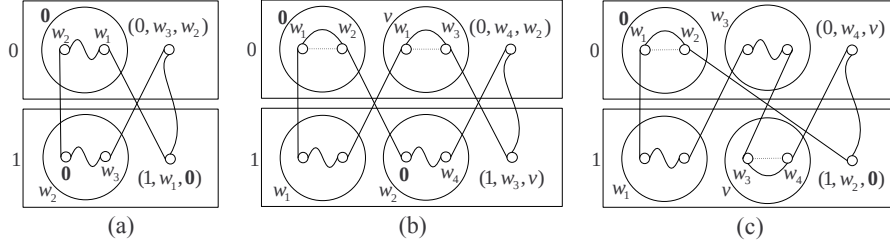


Figure 9: Illustrations of Lemma 6

$CQ_3(0, v) - F(CQ_3(0, v))$. Assume that $C_1 = \langle (0, \mathbf{0}, w_1) \xrightarrow{H_{CQ_3}} (0, \mathbf{0}, w_2) \rightarrow_{CQ_3} (0, \mathbf{0}, w_1) \rangle$ and $C_2 = \langle (0, v, w_1) \xrightarrow{H_{CQ_3}} (0, v, w_3) \rightarrow_{CQ_3} (0, v, w_1) \rangle$. (see fig.9(b)) By Observation 1, there exists a path $P((0, \mathbf{0}, w_1), (0, v, w_1))$ that it traverses every vertex of $V(CQ_3(1, w_1)) \cup \{(0, \mathbf{0}, w_1), (0, v, w_1)\}$ exactly once. Let $(1, w_2, w_4)$ be a vertex of $CQ_3(1, w_2)$ with $w_4 \neq \mathbf{0}$ and $w_4 \neq v$. Let H be the subgraph of $HCC(1, 3)$ induced by vertices of $HCC(1, 3) - (CQ_3(0, \mathbf{0}) \cup CQ_3(0, v) \cup CQ_3(1, w_1) \cup CQ_3(1, w_2))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((1, w_3, v), (0, w_4, w_2))$ in H . Let $C = \langle P((0, \mathbf{0}, w_1), (0, v, w_1)) \xrightarrow{H_{CQ_3}} (0, v, w_3) \rightarrow_{Q_1} P((1, w_3, v), (0, w_4, w_2)) \rightarrow_{Q_1} (1, w_2, w_4) \xrightarrow{H_{CQ_3}} (1, w_2, \mathbf{0}) \rightarrow_{Q_1} (0, \mathbf{0}, w_2) \xrightarrow{H_{CQ_3}} (0, \mathbf{0}, w_1) \rangle$. Then C forms a hamiltonian cycle of $HCC(1, 3) - F$.

Subcase 2.1.2: $u = 1$.

Since CQ_3 is 1-fault hamiltonian, there exists a hamiltonian cycle C_1 in $CQ_3(0, \mathbf{0}) - F(CQ_3(0, \mathbf{0}))$. Let $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ be two successive vertices of C_1 with $w_1 \neq v$ and $w_2 \neq v$. (see fig.9(c)) Similarly, there exists a hamiltonian cycle C_2 in $CQ_3(1, v) - F(CQ_3(1, v))$. Let $(1, v, w_3)$ and $(1, v, w_4)$ be two successive vertices of C_2 with $w_3 \neq \mathbf{0}$ and $w_4 \neq \mathbf{0}$. By Observation 2, since $w_1 \neq v$ and $w_3 \neq \mathbf{0}$, there exists a path $P((0, \mathbf{0}, w_1), (1, v, w_3))$ that it traverses every vertex of $V(CQ_3(1, w_1)) \cup V(CQ_3(0, w_3)) \cup \{(0, \mathbf{0}, w_1), (1, v, w_3)\}$ exactly once. Let H be the subgraph of $HCC(1, 3)$ induced by vertices of $HCC(1, 3) - (CQ_3(0, \mathbf{0}) \cup CQ_3(0, w_3) \cup CQ_3(1, v) \cup CQ_3(1, w_1))$. By Lemma 4, since $|F(H)| = 0$, there exists a hamiltonian path $P((0, w_4, v), (1, w_2, \mathbf{0}))$ in H . Let $C = \langle P((0, \mathbf{0}, w_1), (1, v, w_3)) \xrightarrow{H_{CQ_3}} (1, v, w_4) \rightarrow_{Q_1} P((0, w_4, v), (1, w_2, \mathbf{0})) \rightarrow_{Q_1} (0, \mathbf{0}, w_2) \xrightarrow{H_{CQ_3}} (0, \mathbf{0}, w_1) \rangle$. Then C forms a hamiltonian cycle of $HCC(1, 3) - F$.

Subcase 2.2: $n > 3$ or $F \neq F_v \cup F_{int}$.

Since CQ_n is $(n-2)$ -fault hamiltonian, there exists a hamiltonian cycle C_1 of $CQ_n(0, \mathbf{0}) - F(CQ_n(0, \mathbf{0}))$. Since $|F - F(CQ_n(0, \mathbf{0}))| = 1$, it is not difficult to find two successive

vertices $(0, \mathbf{0}, w_1)$ and $(0, \mathbf{0}, w_2)$ of C_1 such that external edges $((0, \mathbf{0}, w_1), (1, w_1, \mathbf{0}))$ and $((0, \mathbf{0}, w_2), (1, w_2, \mathbf{0}))$ are both fault free. (see fig.9(a)) Let $((1, w_2, w_3), (0, w_3, w_2))$ be a fault free external edge of $HCC(1, n)$ with $w_3 \neq \mathbf{0}$ and let H be the subgraph of $HCC(1, n)$ that it be induced by vertices of $HCC(1, n) - (CQ_n(0, \mathbf{0}) \cup CQ_n(1, w_2))$. Suppose that $n = 3$ and $F \neq F_v \cup F_{int}$. Since $|F| = n - 1 = 2$ and $|F(CQ_3(0, \mathbf{0}))| = n - 2 = 1$, $|F_{ext}| = 1$. That is, every $CQ_3(i, j)$ of $HCC(1, 3)$ except $CQ_3(0, \mathbf{0})$ satisfies $|F(CQ_3(i, j))| = 0$. Suppose that $n > 3$. Since $|F(CQ_n(0, \mathbf{0}))| = n - 2$, every $CQ_n(i, j)$ of $HCC(1, n)$ except $CQ_n(0, \mathbf{0})$ satisfies $|F(CQ_n(i, j))| \leq n - 3$. In addition, $|F(H)| \leq 1 \leq 2^n - 6$. By lemma 4, there exists a hamiltonian path $P((0, w_3, w_2), (1, w_1, \mathbf{0}))$ in $H - F(H)$. Let $C = \langle (0, \mathbf{0}, w_1) \xrightarrow{H_{CQ_n}} (0, \mathbf{0}, w_2) \rightarrow_{Q_1} (1, w_2, \mathbf{0}) \xrightarrow{H_{CQ_n}} (1, w_2, w_3) \rightarrow_{Q_1} P((0, w_3, w_2), (1, w_1, \mathbf{0})) \rightarrow_{Q_1} (0, \mathbf{0}, w_1) \rangle$. Then C forms a hamiltonian cycle of $HCC(1, n) - F$.

Case 3: $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$.

Let $((u, v, w), (\bar{u}, w, v))$ be a fault free external edge in $HCC(1, n)$. Let $t = 2^n$. By lemma 4, since $|F| \leq n - 1 \leq t - 4 = 2^n - 4$ and $|F(CQ_n(i, j))| \leq n - 3$ for each $CQ_n(i, j)$, there exists a hamiltonian path $P((u, v, w), (\bar{u}, w, v))$ in $HCC(1, n) - F$. Let $C = \langle P((u, v, w), (\bar{u}, w, v)) \rightarrow_{Q_1} (u, v, w) \rangle$. Then C forms a hamiltonian cycle of $HCC(1, n) - F$. This lemma is proved. $\square$

For $i = 0, 1$, $k \geq 1$, and $n \geq 1$, denote by $H_{k-1}(i)$ the subgraph of $HCC(k, n)$ induced by the vertices of $\{(iu, v, w) : u \in \{0, 1\}^{k-1}, v, w \in \{0, 1\}^n\}$. Clearly, $H_{k-1}(0)$ and $H_{k-1}(1)$ are isomorphic to $HCC(k-1, n)$. Let $x = (iu_1, v_1, w_1) \in H_{k-1}(i)$ and $x'' = (\bar{i}u_1, w_1, v_1)$ to be the neighbor of x in $H_{k-1}(\bar{i})$.

Lemma 7 Let $k \geq 2$ and $n \geq 3$. Suppose that $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, $(k+n-4)$ -fault hamiltonian connected. Then $HCC(k, n)$ is $(k+n-3)$ -fault hamiltonian connected.

Proof. Let F be the fault set of $HCC(k, n)$ with $|F| \leq k + n - 3$. Without loss of generality, assume that $|F(H_{k-1}(0))| \geq |F(H_{k-1}(1))|$. Let x and y be two fault free vertices in $HCC(k, n)$. We want to construct a hamiltonian path of $HCC(k, n) - F$ joining x and y .

Case 1: $|F(H_{k-1}(0))| \leq k + n - 4$ and $|F(H_{k-1}(1))| \leq k + n - 4$.

Subcase 1.1: $x, y \in H_{k-1}(0)$ or $x, y \in H_{k-1}(1)$.

Without loss of generality, assume that $x, y \in H_{k-1}(0)$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-4)$ -fault hamiltonian connected, there exists a hamiltonian path P of $H_{k-1}(0) - F(H_{k-1}(0))$ joining x and y . Since $\lfloor l(P)/2 \rfloor > |F - F(H_{k-1}(0))|$, there exists two successive vertices z_1 and z_2 on P such that the external edges (z_1, z_1'') and (z_2, z_2'') are both fault free. Then we write path P as $\langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_2 \rightarrow P_2 \rightarrow y \rangle$. (see fig.10(a)) Similarly, there exists a hamiltonian path P_3 of $H_{k-1}(1) - F(H_{k-1}(1))$ joining z_1'' and z_2'' . Then $\langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_1'' \rightarrow P_3 \rightarrow z_2'' \rightarrow z_2 \rightarrow P_2 \rightarrow y \rangle$ forms a hamiltonian path of $HCC(k, n) - F$ joining x and y .

Subcase 1.2: $x \in H_{k-1}(0)$ and $y \in H_{k-1}(1)$.

Let z_1 be the vertex of $H_{k-1}(0)$ such that the external edge (z_1, z_1'') is fault free with $z_1 \neq x$ and $z_1'' \neq y$. (see fig.10(b)) By hypothesis, since $HCC(k-1, n)$ is $(k+n-4)$ -fault hamiltonian connected, there exist a hamiltonian path P_1 of $H_{k-1}(0) - F(H_{k-1}(0))$ joining x and z_1 and a hamiltonian path P_2 of $H_{k-1}(1) - F(H_{k-1}(1))$ joining z_1'' and y . Then $\langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_1'' \rightarrow P_2 \rightarrow y \rangle$ forms a hamiltonian path of $HCC(k, n) - F$ joining x and y .

Case 2: $|F(H_{k-1}(0))| = k + n - 3$.

Hence $|F(H_{k-1}(1))| = 0$. Next, consider the following three subcases.

Subcase 2.1: $x, y \in H_{k-1}(0)$.

Let $f_1 \in F(H_{k-1}(0))$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-4)$ -fault hamiltonian connected, there exists a hamiltonian path P of $H_{k-1}(0) - (F(H_{k-1}(0)) - \{f_1\})$ joining x and y . We consider f_1 with two cases: (1) Suppose $f_1 \in P$, let two vertices z_1 and z_2 be the neighbors of f_1 on P and let $P = \langle x \rightarrow P_1 \rightarrow z_1 \rightarrow f_1 \rightarrow z_2 \rightarrow P_2 \rightarrow y \rangle$. (2) Suppose $f_1 \notin P$, let $P = \langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_2 \rightarrow P_2 \rightarrow y \rangle$. (see fig.10(a)) Similarly, there exists a hamiltonian path P_3 of $H_{k-1}(1)$ joining z_1'' and z_2'' . Then $\langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_1'' \rightarrow P_3 \rightarrow z_2'' \rightarrow z_2 \rightarrow P_2 \rightarrow y \rangle$ forms a hamiltonian path of $HCC(k, n) - F$ joining x and y .

Subcase 2.2: $x \in H_{k-1}(0)$ and $y \in H_{k-1}(1)$.

By hypothesis, since $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, there exists a hamiltonian cycle C in $H_{k-1}(0) - F(H_{k-1}(0))$. Let z_1 be a neighbor of x on C with $z_1'' \neq y$ and let $C = \langle x \rightarrow P_1 \rightarrow z_1 \rightarrow x \rangle$. (see fig.10(b)) By hypothesis, since $HCC(k-1, n)$ is hamiltonian connected, there exists a hamiltonian path P_2 of $H_{k-1}(1)$ joining z_1'' and y . Then $\langle x \rightarrow P_1 \rightarrow z_1 \rightarrow z_1'' \rightarrow P_2 \rightarrow y \rangle$ forms a hamiltonian path of $HCC(k, n) - F$ joining x and y .

Subcase 2.3: $x, y \in H_{k-1}(1)$.

We consider x'' and y'' with two subcases.

Subcase 2.3.1: x'' or $y'' \notin F$.

Without loss of generality, assume that $x'' \notin F$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, there exists a hamiltonian cycle C in $H_{k-1}(0) - F(H_{k-1}(0))$. Let z_1 be a neighbor of x'' on C with $z_1'' \neq y$ and let $C = \langle x'' \rightarrow P_1 \rightarrow z_1 \rightarrow x'' \rangle$. (see fig.10(c)) By hypothesis, since $HCC(k-1, n)$ is $(k+n-4)$ -fault hamiltonian connected, there exists a hamiltonian path P_2 of $H_{k-1}(1) - \{x\}$ joining z_1'' and y . Then $\langle x \rightarrow x'' \rightarrow P_1 \rightarrow z_1 \rightarrow z_1'' \rightarrow P_2 \rightarrow y \rangle$ forms a hamiltonian path of $HCC(k, n) - F$ joining x and y .

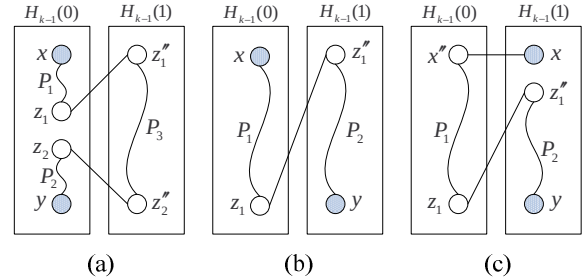


Figure 10: Illustrations of Lemma 7

Subcase 2.3.2: $x'', y'' \in F$.

For $i = 0, 1$, let $H'_{k-1}(i)$ induced by the vertices of $\{(0iu', v, w), (1iu', v, w) : u' \in \{0, 1\}^{k-2}, v, w \in \{0, 1\}^n\}$. We consider the relative positions of x and y with two cases: (1) Suppose $x, y \in H'_{k-1}(0)$ or $x, y \in H'_{k-1}(1)$, without loss of generality, assume that $x, y \in H'_{k-1}(0)$. Then x'', y'' are also in $H'_{k-1}(0)$, e.g., $x'' = (00, 001, 000)$ and $y'' = (00, 100, 110)$ if $x = (10, 000, 001)$ and $y = (10, 110, 100)$. (see fig.11(a)) When $|F(H'_{k-1}(0))| \leq k + n - 4$ and $|F(H'_{k-1}(1))| \leq k + n - 4$, the proof of this case is similar to Subcase 1.1; when $|F(H'_{k-1}(0))| = k + n - 3$, the proof of this case is similar to Subcase 2.1. (2) Suppose $x \in H'_{k-1}(0)$ and $y \in H'_{k-1}(1)$, the vertex x'' is in $H'_{k-1}(0)$ and the vertex y'' is in $H'_{k-1}(1)$, e.g., $x'' = (00, 001, 000)$ and $y'' = (01, 100, 110)$ if $x = (10, 000, 001)$ and $y = (11, 110, 100)$. (see

fig.11(b)) Then $|F(H'_{k-1}(0))| \leq k + n - 4$ and $|F(H'_{k-1}(1))| \leq k + n - 4$, the proof of this case is similar to Subcase 1.2.

This lemma is proved. $\square$

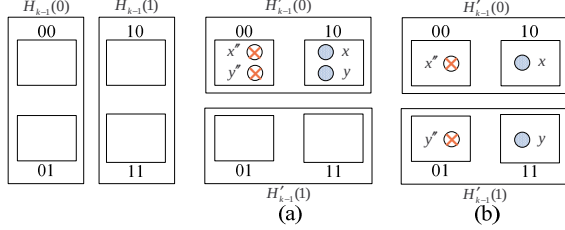


Figure 11: Illustrations of Subcase 2.3.2 in Lemma 7

Lemma 8 Let $k \geq 2$ and $n \geq 3$. Suppose that $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, $(k+n-4)$ -fault hamiltonian connected. Then $HCC(k, n)$ is $(k+n-2)$ -fault hamiltonian.

Proof. Without loss of generality, assume that $|F(H_{k-1}(0))| \geq |F(H_{k-1}(1))|$.

Case 1: $|F(H_{k-1}(0))| = k + n - 2$.

Hence, $|F(H_{k-1}(1))| = 0$. Let $f_1 \in F(H_{k-1}(0))$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, there exists a hamiltonian cycle C in $H_{k-1}(0) - (F(H_{k-1}(0)) - f_1)$. We consider f_1 with two cases: (1) Suppose $f_1 \in C$, let two vertices z_1 and z_2 be the neighbors of f_1 on C and let $C = \langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow f_1 \rightarrow z_1 \rangle$. (2) Suppose $f_1 \notin C$, let $C = \langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow z_1 \rangle$. (see fig.12) By hypothesis, since $HCC(k-1, n)$ is hamiltonian connected, there exists a hamiltonian path P_2 of $H_{k-1}(1)$ joining z'_1 and z'_2 . Then $\langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow z'_2 \rightarrow P_2 \rightarrow z'_1 \rightarrow z_1 \rangle$ forms a hamiltonian cycle of $HCC(k, n) - F$.

Case 2: $|F(H_{k-1}(0))| = k + n - 3$.

Hence, $|F(H_{k-1}(1))| = 1$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-3)$ -fault hamiltonian, there exists a hamiltonian cycle C in $H_{k-1}(0) - F(H_{k-1}(0))$. It is not difficult to verify that there exist two successive vertices z_1 and z_2 on C such that the external edges (z_1, z'_1) and (z_2, z'_2) are both fault free. (see fig.12) Let $C = \langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow z_1 \rangle$. By hypothesis, since $HCC(k-1, n)$ is $(k+n-4)$ -fault hamiltonian connected, there exists a hamiltonian path P_2 of $H_{k-1}(1) - F(H_{k-1}(1))$ joining z'_1 and z'_2 . Then $\langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow z'_2 \rightarrow P_2 \rightarrow z'_1 \rightarrow z_1 \rangle$ forms a hamiltonian cycle of $HCC(k, n) - F$.

Case 3: $|F(H_{k-1}(0))| \leq k + n - 4$ and $|F(H_{k-1}(1))| \leq k + n - 4$.

Let z_1 and z_2 be two vertices of $H_{k-1}(0)$ that the external edges (z_1, z'_1) and (z_2, z'_2) are both fault free. By hypothesis, since $HCC(k-1, n)$ is

$(k+n-4)$ -fault hamiltonian connected, there exist a hamiltonian path P_1 of $H_{k-1}(0) - F(H_{k-1}(0))$ joining z_1 and z_2 and a hamiltonian path P_2 of $H_{k-1}(1) - F(H_{k-1}(1))$ joining z'_1 and z'_2 . (see fig.12) Then $\langle z_1 \rightarrow P_1 \rightarrow z_2 \rightarrow z'_2 \rightarrow P_2 \rightarrow z'_1 \rightarrow z_1 \rangle$ forms a hamiltonian cycle of $HCC(k, n) - F$. This lemma is proved. $\square$

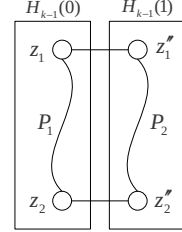


Figure 12: Illustration of Lemma 8

Finally, we provide the main results for $HCC(k, n)$ as Theorem 1. By the similar proof techniques in Lemma 5 to Lemma 8, we also can prove Theorem 2.

Theorem 1 Let $k \geq 1$ and $n \geq 3$, $HCC(k, n)$ is $(k+n-2)$ -fault hamiltonian and $(k+n-3)$ -fault hamiltonian connected.

Theorem 2 Let $k \geq 1$ and $n \geq 3$, $HRHL(k, n)$ is $(k+n-2)$ -fault hamiltonian and $(k+n-3)$ -fault hamiltonian connected.

4 Conclusion

In this paper, we show that $HCC(k, n)$ is $(k+n-2)$ -fault hamiltonian and $(k+n-3)$ -fault hamiltonian connected for $k \geq 1$ and $n \geq 3$. We can use the same way to prove that the other members of hierarchical restricted HL-graphs. Hence, $HRHL(k, n)$ is $(k+n-2)$ -fault hamiltonian and $(k+n-3)$ -fault hamiltonian connected for $k \geq 1$ and $n \geq 3$.

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